A.9. Additional latent visualizations from linear probes

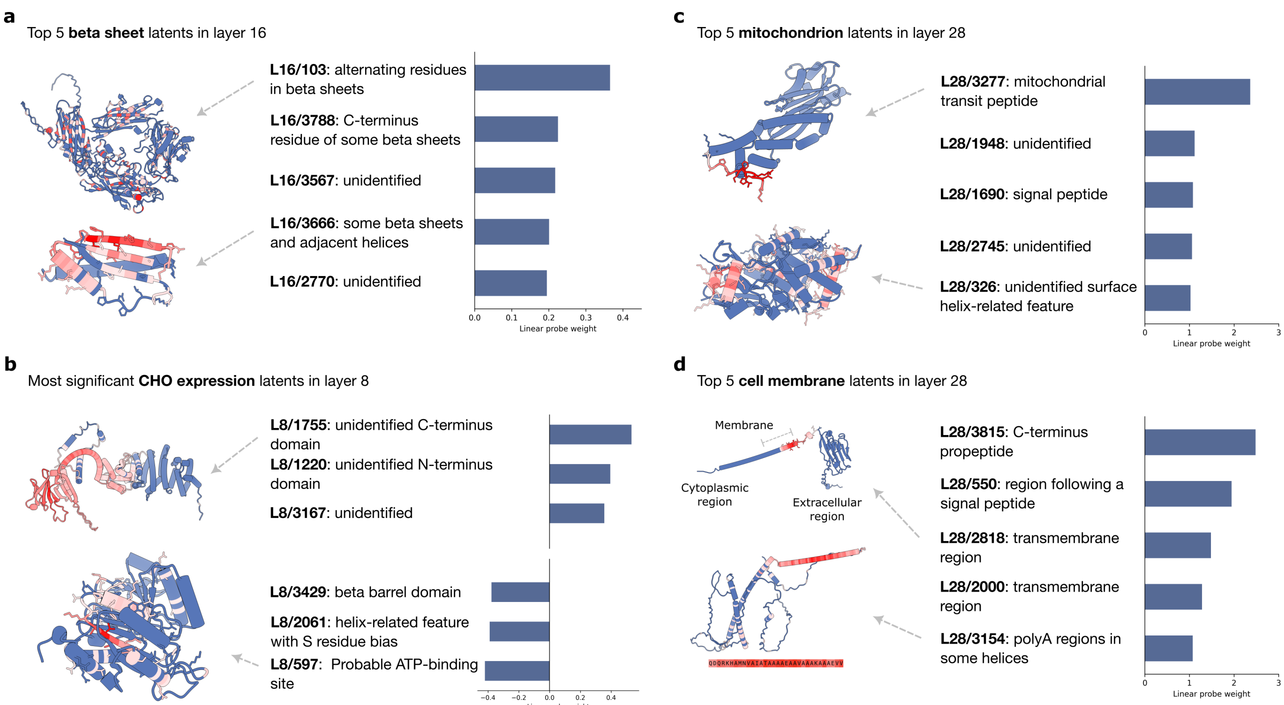

Figure 9. Linear probes identify predictive SAE latents for downstream tasks. **a**. The top 5 beta sheet latents from the secondary structure prediction task (SAE trained on ESM layer 16). **b**. The latents corresponding to the 3 most positive and 3 most negative coefficients from the linear probe on CHO cell line expression. The most positive latent activates on an unidentified motif; the most negative latent activates on an ATP binding site (SAE trained on ESM layer 8). **c-d**. Top 5 mitochondrion and cell membrane latents from the subcellular localization prediction task. The top nucleus latents identify transit peptides. The top cell membrane latents correlate with signal peptides and transmembrane regions (SAE trained on ESM layer 28).

A.10. Distribution of linear probe coefficients

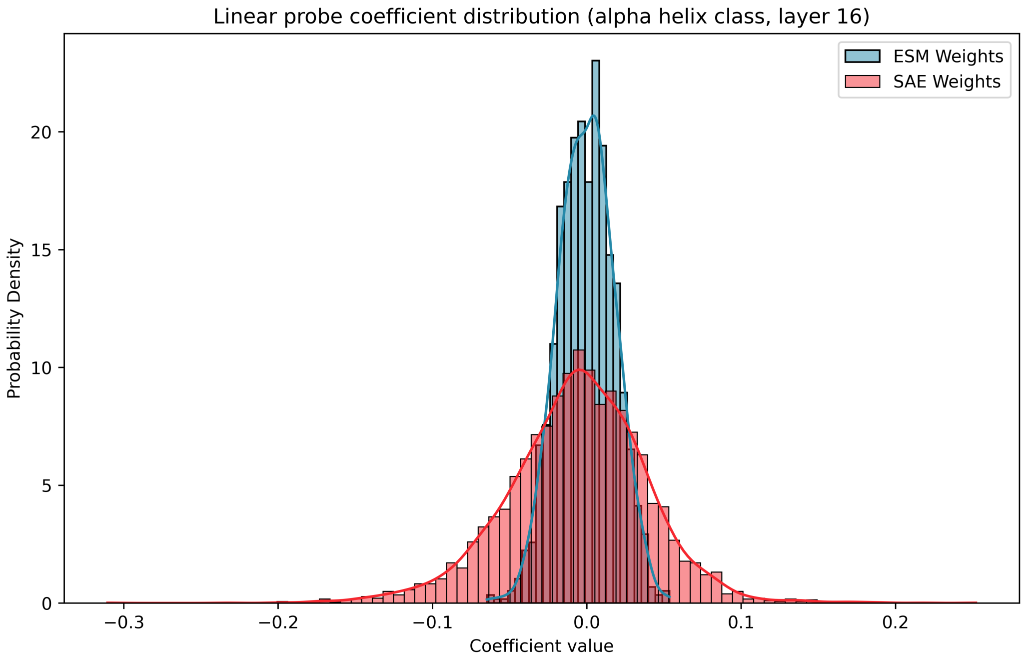

Figure 10. Example distribution of ESM vs. SAE linear probe coefficients. Shown here are coefficients for the alpha helix class from the secondary structure prediction task, using layer 16. The SAE coefficients have higher variance ($2.03\mathrm{e}{-3}$) than the ESM coefficients ($3.24\mathrm{e}{-4}$) because more latents are individually predictive of the label. The SAE coefficients also have a larger negative skew ($-2.05\mathrm{e}{-1}$) compared to ESM ($-6.51\mathrm{e}{-2}$). This can be explained by the SAE latents being more monosemantic: they sort more distinctly into helix vs. non-helix classes, and there are more non-helix examples than helix ones.